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Waveform Analysis for Nuclear Astrophysics Experiments with TACTIC

Research Project Report

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Abstract

This report presents the results of a summer research project carried out at the University of York. The project focused on the analysis of waveform data acquired using a ^{241}Am radiation source detected with a gaseous proportional detector. A planar analogue of the TACTIC detector, the test chamber, was used. It implements the μ -RWELL detector technology for position-sensitive charge readout. Several waveform characteristics, including multiplicity, pile-up and baseline offsets, were investigated, with particular emphasis on pulse amplitude. ROOT-based analysis routines were developed to process, visualise and extract information from the acquired waveforms. A preliminary comparison of two baseline-estimation methods yielded a 3.43% relative difference in the estimated amplitude resolution. Furthermore, an anomalous detector response to the radiation source was identified and investigated, and its origin attributed to a time-dependent charge-up effect in the μ -RWELL. The charge-up interpretation was also connected with the non-uniform irradiation produced by the absence of a collimator.

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Chapter 1

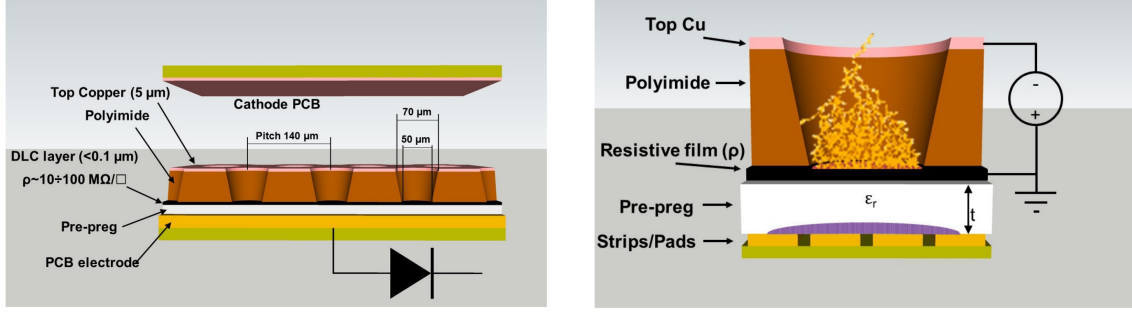
Introduction

1.1 Research context

In the context of low-energy nuclear physics, active-target detectors provide an effective approach for studying nuclear reactions. One example is TACTIC (TRIUMF Annular Chamber for Tracking and Identification of Charged particles), a cylindrical active-target time projection chamber (TPC) that has been developed and studied at the University of York over the past few years [1, 2]. Its main purpose is the measurement of cross sections for alpha-induced nuclear reactions at Canada’s national particle accelerator centre, TRIUMF. Inside it, the fill gas acts as both the target and detection medium, whereas the μ -RWELL amplification stage provides sensitivity over a wide range of deposited energies. Its segmented cylindrical geometry enables trajectory reconstruction, and the cathode cage feature suppresses ionisation from the unreacted beam to reduce pile-up [1, 2]. Characterisation of this detector requires the study of many aspects of its operation, a small fraction of them being the main research focus of this report.

1.2 The μ -RWELL

The μ -RWELL (micro-Resistive WELL) is a compact MPGD (Micro-Pattern Gaseous Detector) developed for position-sensitive particle detection [3, 2]. The complete detector assembly comprises a cathode PCB and a μ -RWELL PCB, the latter containing the amplification and readout structure. As shown in Fig. 1.1a, the μ -RWELL PCB mainly includes a copper-clad polyimide foil patterned with microscopic blind holes, a Diamond-Like Carbon (DLC) resistive layer and a segmented readout PCB [3, 1]. The cathode is held at a negative potential, and ionisation electrons are guided into the holes by the drift field applied between the cathode and the top copper electrode [3].



(a) Layout of the μ -RWELL featuring the different layers.

(b) Structure of a hole with an electron avalanche to show the working principle of the detector.

Figure 1.1: Illustrations of the μ -RWELL detector showing its structure and working principle. Reproduced from Bencivenni et al. [3].

Upon applying a potential difference between the top copper electrode and the resistive layer, electrons entering the holes undergo avalanche multiplication, much like in a GEM¹. In this report, this potential difference across the μ -RWELL is referred to as the *mesh voltage*. The avalanche charge is then collected at the resistive layer, which is coupled to the segmented anode pads underneath, where the readout signal is induced [1]. The DLC layer allows for current evacuation while producing a local voltage drop during a discharge, effectively suppressing destructive sparks.

1.3 The test chamber

Before measurements are performed with TACTIC, different aspects of the detection process can be studied with its planar analogue, the test chamber. The drift field, in this case, is applied between a lower cathode plate and the μ -RWELL, situated at the upper plate of the chamber. Its planar geometry implies that the electric field is approximately constant over the drift region, allowing for easier analysis of the properties of the detecting mechanism [1].

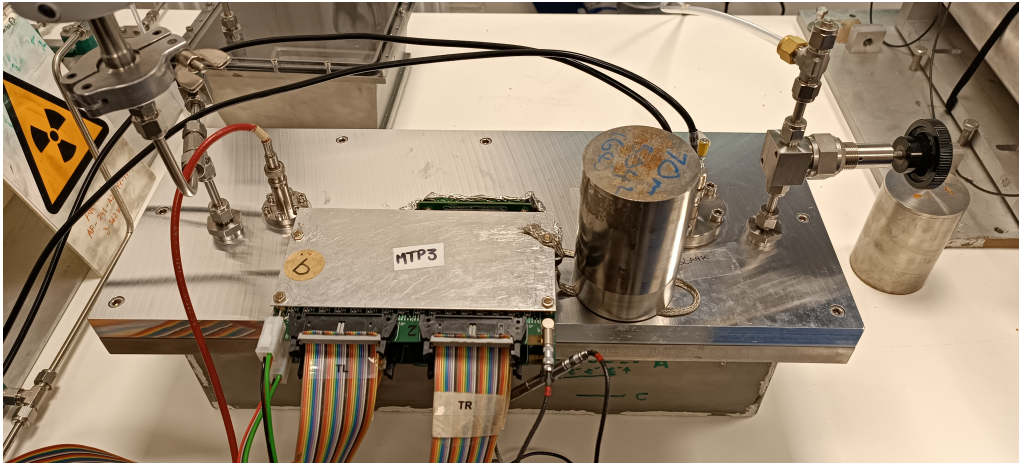


Figure 1.2: Picture of the test chamber in the lab.

¹A Gas Electron Multiplier (GEM) is a micro-pattern gaseous detector in which avalanche multiplication occurs within microscopic holes to detect ionising radiation [4].

In the lab setup used in this work, the μ -RWELL was segmented into 60 pads, 4.1 mm wide each, with a spacing of 0.1 mm. To a first approximation, ionisation electrons drift perpendicular to the readout plane, so the index of the pad at which a signal was detected can give information about the trajectory of the particle [1].

A useful schematic of the interaction process inside the chamber is drawn in Fig. 1.3, which also shows the position of the radiation source. In this study, a ^{241}Am source was used, whose dominant alpha line has an energy of 5.486 MeV [5].

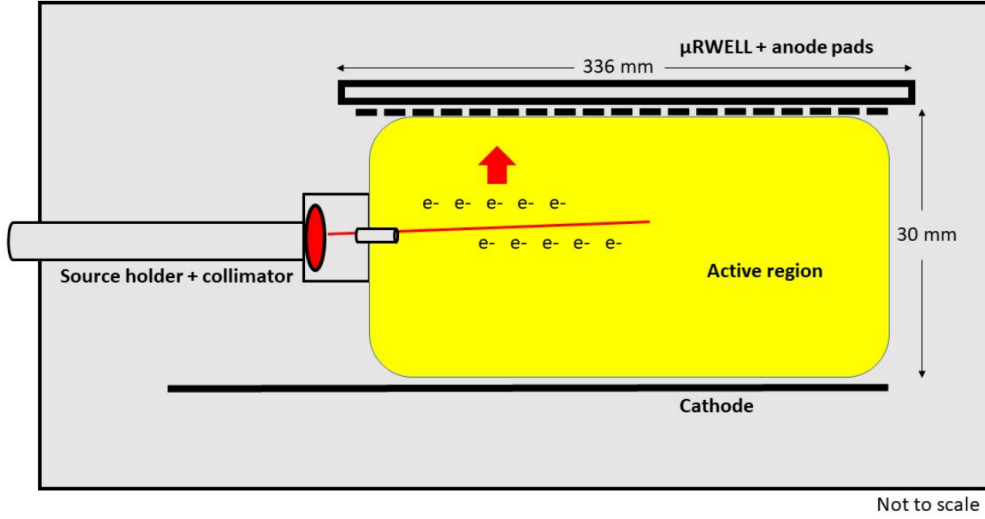


Figure 1.3: Illustration of the working mechanism of the test chamber [1].

1.4 The readout chain

As seen in Fig. 1.2, two 32-channel charge-sensitive preamplifiers were used to read out the 60 anode pads, leaving four channels unconnected [1]. Their job is to convert the charge signal induced on each pad into voltage pulses that the digitiser can subsequently record.

The digitiser², which is connected to the preamplifiers via flat ribbon cables, is responsible for converting continuous analogue voltage from the preamplifiers into numerical samples. Finally, these samples are processed by the data acquisition system (DAQ) and stored as events in data files. In the datasets analysed here, each event contained waveforms of 255 samples per channel.

An event was recorded every time the configured trigger condition was satisfied [6]. In this case, the trigger condition was for any sample in any of the channels to fall under a set threshold. Ideally, every event would correspond to a pulse. It is common, however, that some of the events do not contain a genuine detector pulse, but rather electronic noise that has crossed the trigger threshold or other types of pulses that will be discussed later.³

²The digitiser used in this setup was a CAEN V1740 [6].

³Here, amplitude denotes the difference between the measured ADC value and the estimated baseline.

1.5 Objectives

As an alpha particle traverses the gas, it loses energy primarily through ionisation and excitation. Its stopping power, $-dE/dx$, depends on the particle velocity and the medium; as the alpha particle slows down, the stopping power generally increases towards the end of its range, so the ionisation density varies along the track [7]. This causes the pulses detected at different anode pads to have different amplitudes. Furthermore, the particle is emitted from the source at an angle, so ionisation electrons produced at different positions travel different distances to the detector. These different drift distances lead to different electron drift times and therefore affect the relative pulse start times [1], and can affect the rise times as well.

On the other hand, characteristics of the detector setup such as bias voltages, gas mixture or microscopic defects in the design may also affect the pulse shapes, amplitudes, range and timing. The aim of this project was to study all these effects by taking different datasets in the lab and subsequently analysing them.

Chapter 2

Waveform analysis and comparison

2.1 Pulser calibration

Before each data-taking sequence in this work, a calibration run was performed to characterise the individual channel response of the preamplifiers under the conditions of the setup at that moment. This was achieved through test pulse injection, using a test pulse generator, or pulser [1].

The available calibration procedure did not produce a consistent channel-by-channel correction and was therefore not applied to the final analysis. This was nevertheless not an obstacle in my subsequent work, as explained later in this report.

2.2 Experimental methods

After the previous step, data from the ^{241}Am source were taken. For that purpose, the electronic components were connected according to the required experimental configuration. A vacuum was then generated inside the chamber, which was subsequently filled with the desired gas mixture in static-fill mode. Afterwards, a bias voltage was applied and the trigger threshold was set. Data were then recorded for approximately five minutes.

The first dataset was recorded with the chamber filled with an Ar:CO₂ mixture at a 90:10 ratio, at 1 bar of pressure. Here, CO₂ acts as a quench gas, absorbing excitation and ultraviolet-photon energy while suppressing photon-induced secondary ionisation and discharge propagation [7, 2]. The characteristics of all five runs are summarised in Table 2.1.

Table 2.1: Experimental data acquired on 8 July 2026 with an Ar:CO₂ mixture at a 90:10 ratio and 1 bar of pressure.

Run	Cathode voltage (V)	Mesh voltage (V)	No. of events
1	-1000	-300	162656
2	-1000	-300	157541
3	-1700	-300	17902
4	-1200	-300	151403
5	-1400	-300	153449

It is worth mentioning that Run 3 has substantially fewer events than the other runs.

This was suspected to have resulted from a rapid change in drift voltage between Runs 2 and 3, which may have caused a high-voltage channel to trip.

2.3 Pulse shape visualisation

Visualising the waveform shapes was an important first step in characterising the recorded signals. Therefore, one of my first steps in this project was to develop a program that could read the data and output a visual representation of them. One of the most convenient and specialised data analysis frameworks for this is ROOT, which is based on C++ and was used throughout this project [8].¹

No existing analysis routine was available for visualising these data, so, after learning how to handle them and making the display more appealing, I plotted the waveforms as seen in Fig. 2.1a. Because this project did not focus quantitatively on the timing aspects of the pulses, the horizontal axis is expressed in sample number rather than physical time. The fact, however, that each waveform exhibits a different baseline offset makes it hard to compare their amplitudes and shapes. To correct that, I subtracted the estimated baseline from each waveform, as shown in Fig. 2.1b. This representation allows for simple comparison of the relative pulse start times, which contain information about the particle trajectory [1].

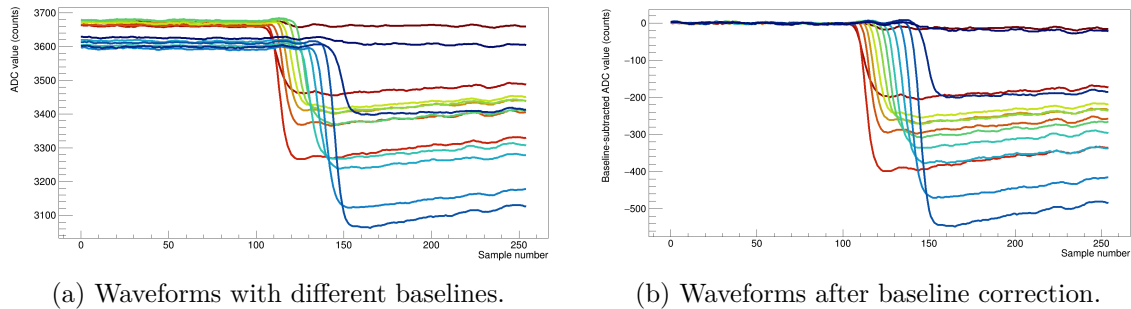


Figure 2.1: Waveform representations of the same event, with and without baseline correction.

Initially, I took the baseline as the value of the first sample of each waveform, which was then subtracted from the whole pulse. Later on, I changed the baseline to be the average of the first 90 samples², providing a less noise-sensitive estimate of the baseline level. Subsequent tests indicated that this change had only a small effect on the subsequent analysis.

I then found that another useful way to represent them was in separate plots, as in Fig. 2.2. This makes it easier to see the individual features of each one of the pulses and the evolution of the alpha particle.

¹ROOT was developed at CERN and is optimised for fields such as particle, nuclear and high-energy physics and the immense amounts of data that they require.

²The first 90 samples had been seen to precede the pulse onset in all events inspected, effectively belonging to the baseline.

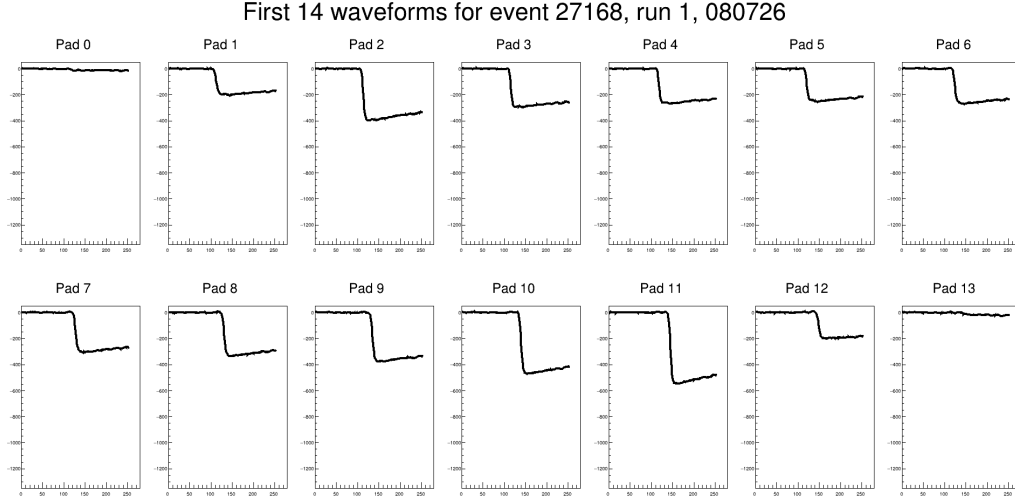


Figure 2.2: Waveforms from the same event displayed individually.

2.4 Baseline correction

To test the impact that a different calculation of the baseline had, I calculated an approximate relative amplitude resolution using the two aforementioned methods. To do that, I represented the differential pulse-height distribution of the summed amplitudes over relevant channels using a histogram [7]. For an approximately Gaussian peak, the distribution can be fitted with a Gaussian function. One can calculate the FWHM (full width at half maximum) of the curve using:

$$FWHM = 2.355 \cdot \sigma, \quad (2.1)$$

where σ is the standard deviation of the curve. The relative resolution as a percentage will then be:

$$R = 100 \cdot \frac{FWHM}{x_0}, \quad (2.2)$$

where x_0 is the peak position.

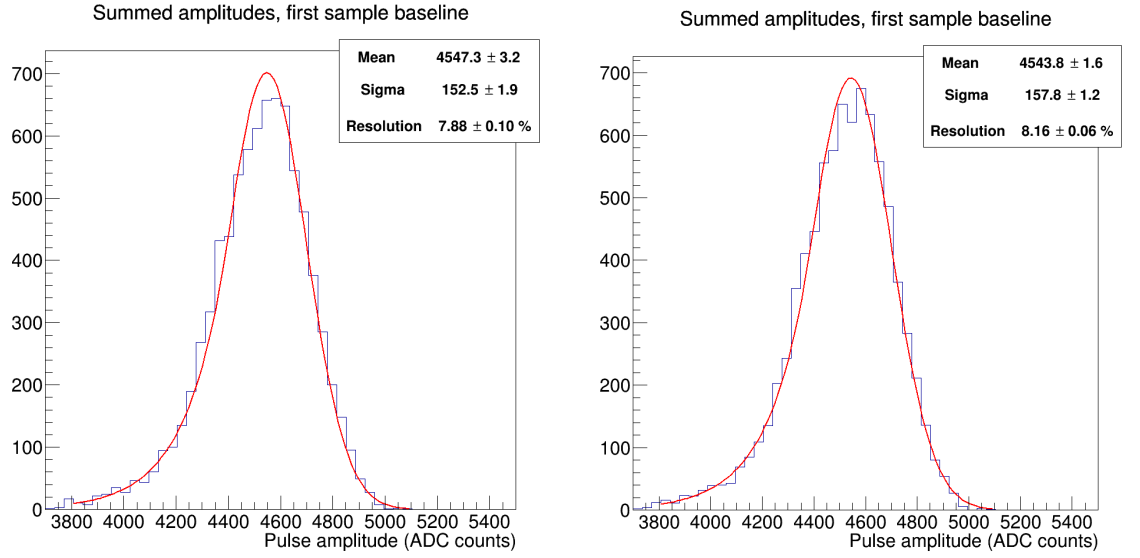
For the two cases, I used exactly the same events to prevent differences in event selection from biasing the comparison. Upon drawing the histogram, however, I observed an asymmetric distribution with a low-amplitude tail, qualitatively resembling a Crystal Ball distribution [9]. The tail was tentatively attributed to different energy-loss mechanisms like incomplete charge collection and incomplete tracks hitting the detector, admitted by the same multiplicity cut. In particular, events with endpoint signals close to the selection threshold can reduce the summed measured amplitude. I therefore fitted the histogram with a Crystal Ball function and used the width of its Gaussian core to define an approximate Gaussian-core relative amplitude resolution:

$$R_{\text{core}} = 100 \cdot \frac{2.355\sigma}{\mu}, \quad (2.3)$$

where σ and μ are the Gaussian-core width and location parameters of the fit, respectively. Because the analysis sums pulse maxima rather than calibrated integrated

charge, the summed amplitude was used only as an energy-related estimator and not as a calibrated measurement of deposited energy.

Fig. 2.3 shows the result. The estimated Gaussian-core relative amplitude resolutions were $7.88 \pm 0.10\%$ with the baseline taken as the first sample and $8.16 \pm 0.06\%$ with the baseline estimated from the average. This corresponds to a 3.43% increase from the first result to the second, so the relative change was small for the purposes of the subsequent analysis.



(a) Histogram taking the baseline as the first sample.

(b) Histogram taking the baseline as the average of the first 90 readings.

Figure 2.3: Comparison of the summed-amplitude distributions for multiplicity-10 events in Run 1 (8 July 2026), using the two baseline-estimation methods.

It is important to note that this was a preliminary estimate. The estimate did not account for correlations and used an approximate empirical fit to an asymmetric distribution. A detailed characterisation of the detector resolution has been carried out in the past and was beyond the scope of this project [1].

2.5 Event selection and rejection

Several event classes were identified and excluded because they did not correspond to isolated alpha-particle events.

2.5.1 Pulse pile-up

When two alpha particles enter the drift region close enough in time, a second pulse may be detected before the detector has had time to return to the baseline after the first one. This phenomenon is known as pulse pile-up [7], and such events were excluded before comparing alpha events because their waveform characteristics differ from those of isolated pulses. Fig. 2.4 shows an example of pulse pile-up.

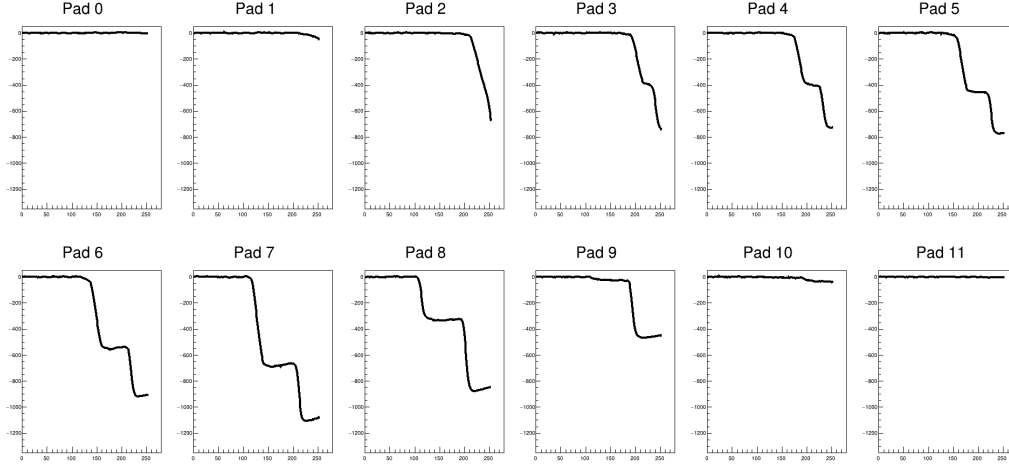


Figure 2.4: Example of an event containing pulse pile-up.

2.5.2 Background interactions

Another class of events excluded from the analysis comprises background interactions. These may originate from interaction of the gas with cosmic rays, the small fraction of gamma rays that the ^{241}Am radiation source produces [5], or other interactions unrelated to the alpha-particle signal of interest.

Candidate background events were identified because their pulses not only tend to have amplitudes outside the typical range, but are also usually detected in one or two isolated pads, without a coherent progression across neighbouring ones. Fig. 2.5 shows an example of a background interaction.

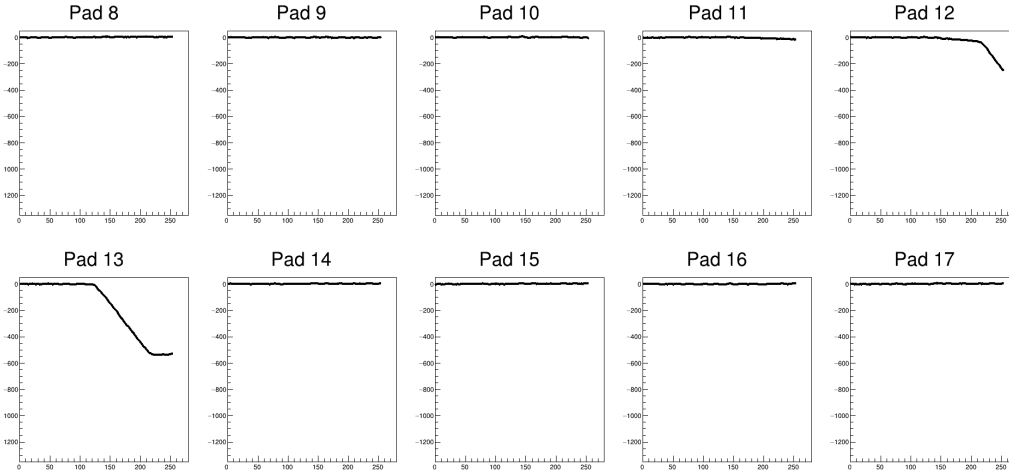


Figure 2.5: Example of an event containing a candidate background interaction.

2.5.3 Noise-triggered events

As stated before, the digitiser may write data that do not correspond to any interaction, but are instead noise that has passed the set threshold by chance. Those cases are easy to identify, since the waveforms consist only of small baseline fluctuations with no identifiable pulse, just like the waveform from pad 0 in Fig. 2.4. These events were therefore excluded

from the pulse-shape analysis.

2.5.4 Multiplicity cut

Event multiplicity was defined as the number of channels whose signal fell below the selected amplitude threshold³. It is therefore a proxy for the track length projected along the segmented direction of the detector [1]. Together with pulse start times, multiplicity also provides information about the angle at which the alpha particle was emitted. Because the ^{241}Am source is dominated by a single alpha line [5], the alpha-particle range is approximately fixed for a given gas condition, and changing the emission angle changes its projection across the pads. Particles with a larger out-of-plane angle therefore tend to cross fewer pads. If the angle is sufficiently large, the track may reach a chamber boundary before the alpha particle stops in the gas, effectively depositing only part of its energy inside the active gas volume.

One way to exclude both empty and background events from the analysis is to apply a multiplicity cut, i.e., rejecting events with multiplicity lower than a set threshold. Imposing some multiplicity also leaves events where the particle travelled rather horizontally, according to our previous arguments. This will not necessarily exclude pulse pile-up, however. For that purpose, one option is to exclude events whose amplitude at a given channel is higher than what would be physically reasonable. This, however, is complicated by the recorded pulse heights being dependent on event multiplicity. Other issues in this aspect of pulse analysis and ways to deal with them are discussed later in this report.

³Recall that pulses are negative with respect to their baseline. This threshold will depend on the noise levels recorded in the lab.

Chapter 3

Pulse amplitude study

As stated before, the heights of the pulses recorded at each channel are expected to be different. A first step towards comparing different events was to plot the pulse height as a function of pad number. It is known that the amplitude of the pulse recorded is proportional to the ionisation energy deposited in the corresponding region [7], provided detector gain and channel response are sufficiently constant. Furthermore, since pad number provides a proxy for position along the track, the previous plot is qualitatively analogous to plotting the local energy deposited over successive pad intervals as a function of distance travelled. As the alpha particle slows, its stopping power increases towards the end of its range, producing a Bragg-like peak in the local energy deposition [7].

3.1 Ar:CO₂ measurements

Unexpectedly, the very first event I tested this for presented an anomaly at pad 2. It is visible in Fig. 3.1 how channel 2 had recorded a much higher amplitude than expected for that pad, which, theoretically, should be slightly lower than that of the following pad. This observation motivated further investigation.

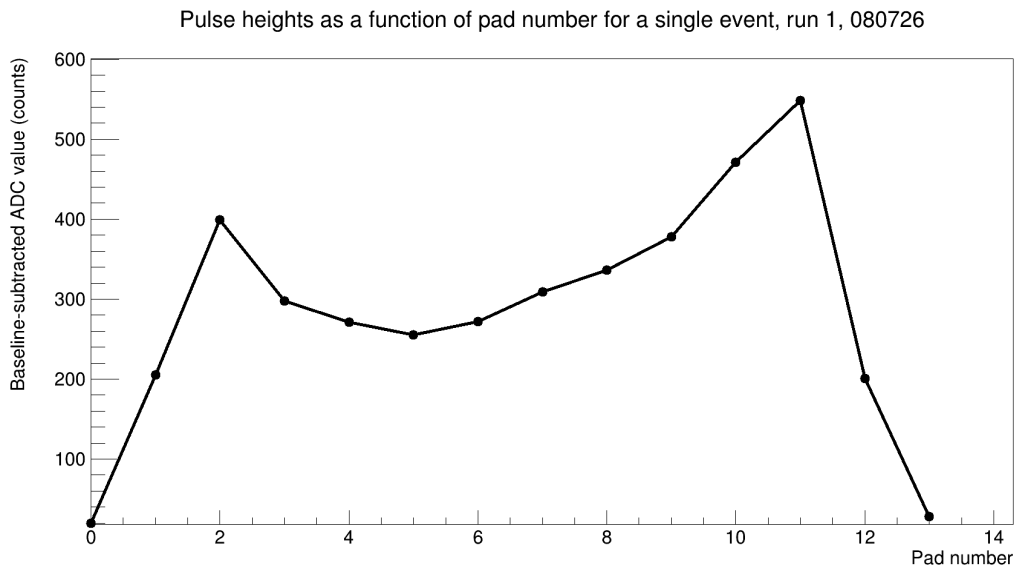


Figure 3.1: Graph of the recorded pulse height as a function of electronic channel number for an event showing an anomalous peak at channel 2.

One of the first explanations I checked was pulse pile-up. The higher amplitude in pad 2 could have been caused by a high-angle alpha that did not reach pad 3, but overlapped at the earlier pads with the high-multiplicity event that was being looked at. This, however, was not the case, since the event being inspected is the one seen in Fig. 2.2. The latter also shows that the anomaly probably was not caused by noise, as the pulse in pad 2 reaches a visibly larger negative amplitude than the one in pad 3, and the observed baseline fluctuations were much smaller than the amplitude difference.

My next step was to determine whether the anomaly persisted across a large number of events. Nevertheless, raw-data analysis initially became a computational bottleneck. I found code optimisation to be insufficient, so I was advised to preprocess the data. I chose ROOT TTree objects, which provide efficient data storage and access, as the right data structure for this [8]. This made the large comparison feasible: 650 million lines of raw data were read, processed, organised and compressed in 1 min 50 s, and the comparison shown in Fig. 3.2 could then be produced in 25 s rather than several hours.

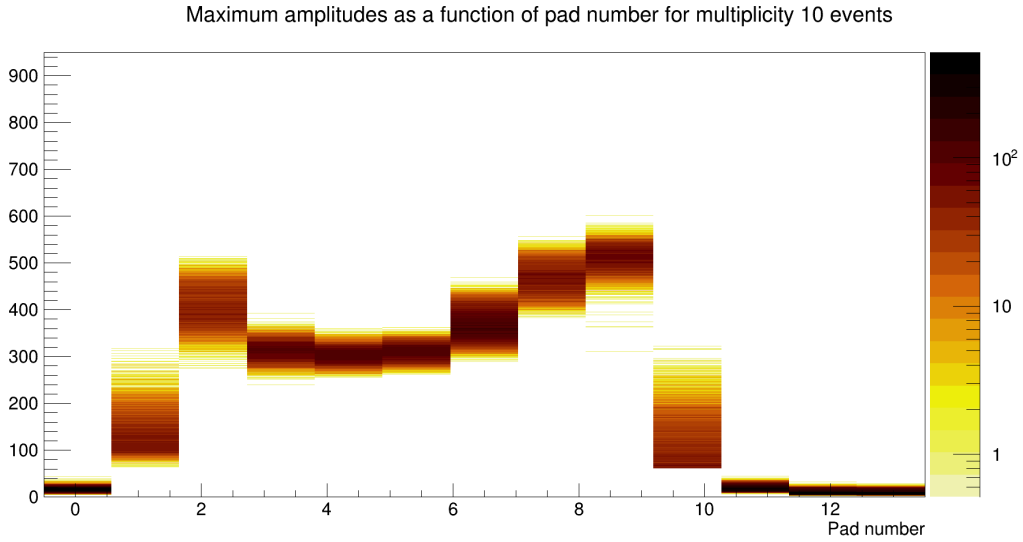


Figure 3.2: 2D histogram for the pulse heights as a function of electronic channel number for many multiplicity-10 events from Run 1 (8 July 2026).

Fig. 3.2 shows the same graph as in Fig. 3.1, this time for all selected multiplicity-10 events simultaneously, represented in a 2D histogram with a suitable colour palette to visualise the event density in each bin. These results showed that pad 2 consistently recorded higher readings than pad 3, a persistence that made a statistical fluctuation unlikely.

As stated before, additional runs had been acquired at different drift fields to study the effect of drift-field strength on the pulses. This was useful to check that the anomaly had partially or completely disappeared for subsequent runs, obtaining the expected Bragg-like profile (Fig. 3.3).

This argued against a fixed preamplifier-channel response as the sole cause of the problem, since the connection was not changed in any way between these two last runs. The absence of an applied pulser correction could still introduce pad-to-pad response differences, but it could not by itself explain the disappearance of the anomaly [1]. It is also apparent that the overall pulse heights in pad numbers higher than 2 are larger in Run

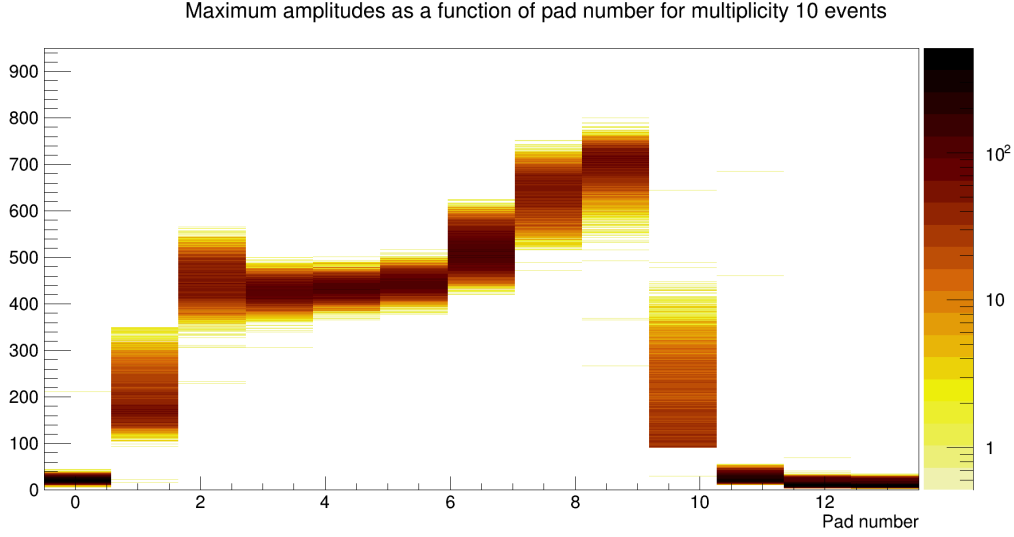
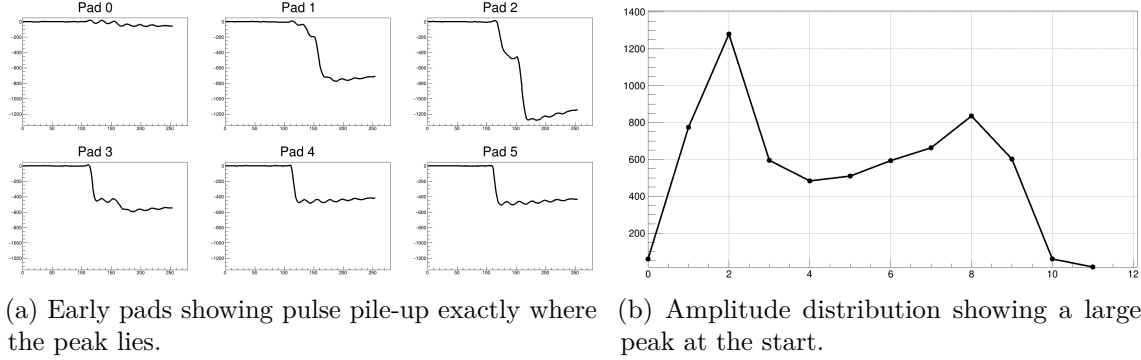


Figure 3.3: 2D histogram for the pulse heights as a function of electronic channel number for many multiplicity-10 events from Run 2 (8 July 2026).

2 than in Run 1, even when the drift field was the same. This was certainly something to note, for which no explanation had yet been identified.

Another observation worth mentioning is that runs with higher drift field¹ presented overall higher amplitudes while the μ -RWELL bias remained unchanged. This is consistent with improved primary-charge collection at higher drift field due to reduced recombination and possible changes in the collection efficiency of electrons into the holes [7, 1].



(a) Early pads showing pulse pile-up exactly where the peak lies.

(b) Amplitude distribution showing a large peak at the start.

Figure 3.4: Waveforms and corresponding amplitude distribution for an anomalous event in Run 4 (8 July 2026).

Even so, there were still events with an unusually high amplitude in pad 2 for all runs. Inspecting those, I found what is pictured in Fig. 3.4a: pulse pile-up. This was consistent with some outlier events, which were readily excluded in this case. This was not the case, however, for most events in Run 1, which showed no double pulses despite exhibiting an anomalous peak at pad 2. I tested this by plotting the waveforms recorded in channel 2 and its neighbours for many events at once, as Fig. 3.5 shows.

¹This is the voltage applied between the cathode and the copper layer of the μ -RWELL. It does not directly set the amplification field inside the wells, which is determined mainly by the μ -RWELL bias.

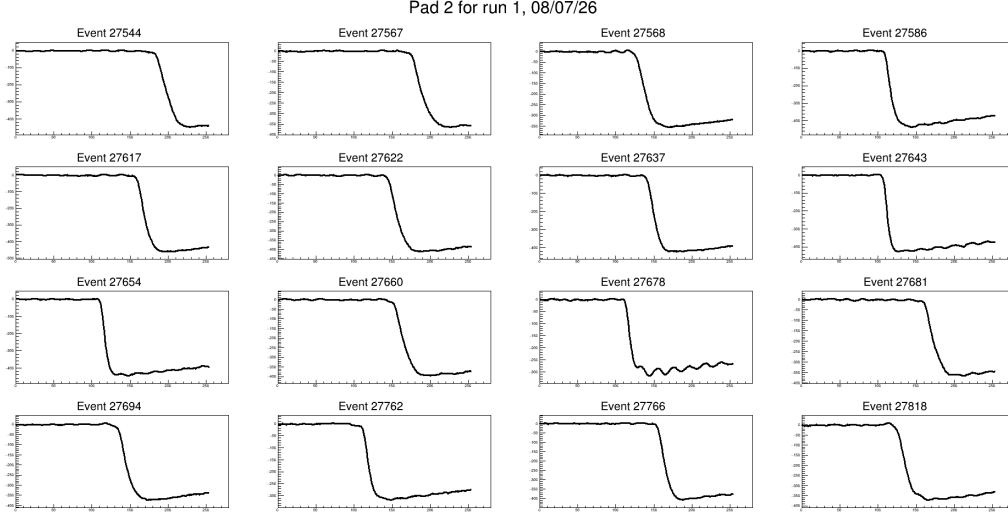


Figure 3.5: Waveforms recorded in channel 2 for several multiplicity-10 events from Run 1 (8 July 2026).

3.2 He:CO₂ measurements

In the absence of a convincing explanation, I examined data acquired on 23 July 2026. The experimental procedure had been the same except that the chamber was instead filled with He:CO₂ at the same ratio, 90:10, again at 1 bar of pressure. The characteristics of the five runs taken are summarised in Table 3.1. Helium is of particular relevance because it serves as the target medium for the TACTIC reactions of interest [1, 2].

Table 3.1: Experimental data acquired on 23 July 2026 with a He:CO₂ mixture at a 90:10 ratio and 1 bar of pressure.

Run	Cathode voltage (V)	Mesh voltage (V)	No. of events
1	-1050	-350	130729
2	-1050	-350	133162
3	-1250	-350	135035
4	-1270	-370	146288
5	-1470	-370	139127

Upon inspection of the amplitudes for Run 1, I identified a channel-to-channel preamplifier connection mismatch (see Fig. 3.6). This suggested a possible electronics-related origin for the initial anomaly. I later rejected that hypothesis for two reasons. On the one hand, the mismatch observed involved the ribbon-cable connection, which carries signals from 16 neighbouring channels, so this particular connection error affected a group of channels rather than only a single pad. On the other hand, the connection in this case was the same as for the previous section, but there was no distinct anomaly in any channel in this case.

This comparison nevertheless required caution. Because helium has a lower linear stopping power for alpha particles than argon [10], the tracks are longer and the deposited energy is spread over a larger distance, tending to reduce the local energy deposition per pad. Moreover, the measured amplitudes also depend on gas-specific quantities such as the

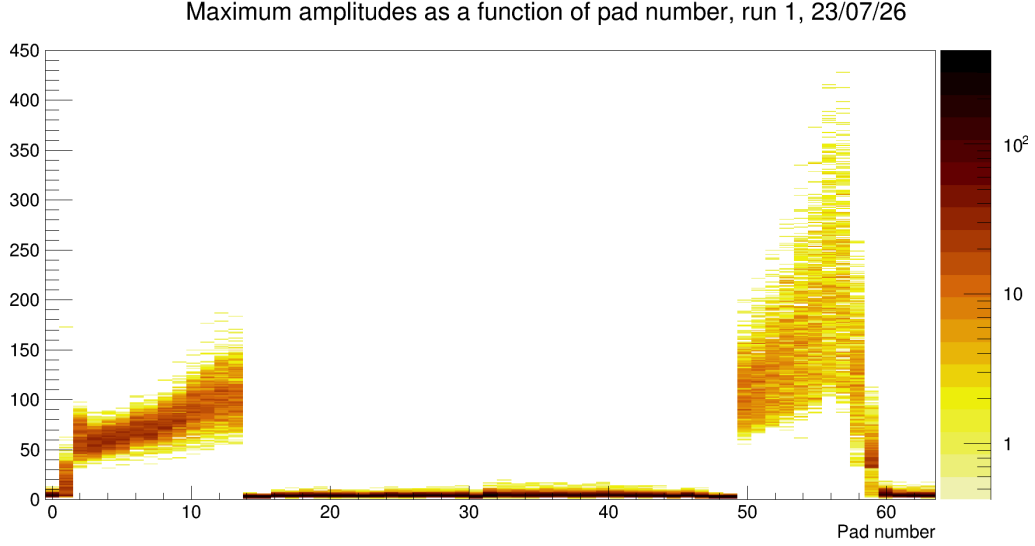


Figure 3.6: Pulse height as a function of electronic channel number for multiplicity-22 events, Run 1 (23 July 2026).

W-value and electron transport and attachment [1]. The Ar- and He-based datasets are therefore not directly comparable, but it was nevertheless worth noting that there was no clear anomaly with helium, even in the first run.

Another hypothesis that had arisen was that the position of the source could have something to do with this. To test that, the source holder was moved back before the second run, yielding the results of Fig. 3.7. Although the position of the Bragg curve changed here, the overall profile remained similar. This argued against fixed microscopic pad-to-pad non-uniformities being the dominant cause of the observed anomaly, even if such variations were not negligible [1]. In that context, we also checked the response of all channels in real time by inspecting the detected pulses while moving the source.

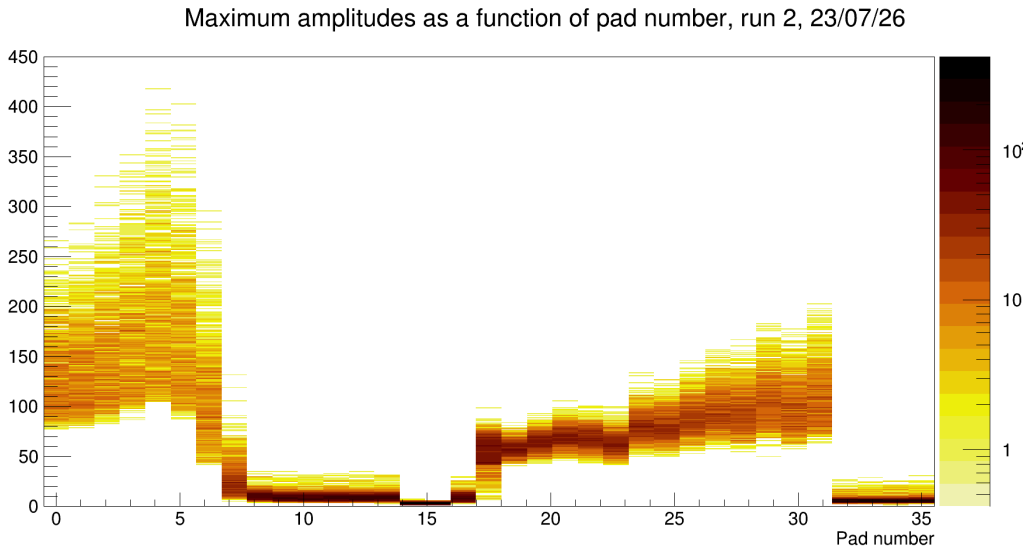


Figure 3.7: Pulse height as a function of electronic channel number for multiplicity-23 events, Run 2 (23 July 2026).

In that sense, the preamplifier mismatch provided an additional validation of the

correct working behaviour of the complete readout chain. The mismatch was subsequently identified and corrected.

Moreover, electronic channels 14, 15, 48 and 49 show near-zero amplitudes. The channel indexing used in these plots includes the four spare preamplifier inputs that are not connected to the μ -RWELL. No readings were lost; all 60 μ -RWELL pads remained connected.

3.3 Preamplifier swap

To assess the influence of the preamplifiers on our measured amplitudes, we swapped them before recording two additional runs of data with an Ar:CO₂ 90:10 mixture at 1 bar. All voltages were the same as in Run 1 from 8 July 2026, the only difference between these two new runs being their acquisition duration and time.

Interestingly, Run 1 did not show a disproportionate peak at the early channels; neither did Run 2. Moreover, the highest amplitudes were almost half of those of Run 1 from 8 July 2026. This was consistent with a possible difference between the gains of the two preamplifiers, as was also observed in the helium data, when comparing the whole range of channels. Furthermore, the spread in amplitudes in channels closer to the Bragg peak was larger, just as with helium data. These results did not allow the contribution of the preamplifiers to the anomaly to be isolated, since some runs acquired with the original preamplifier arrangement also showed no anomaly.

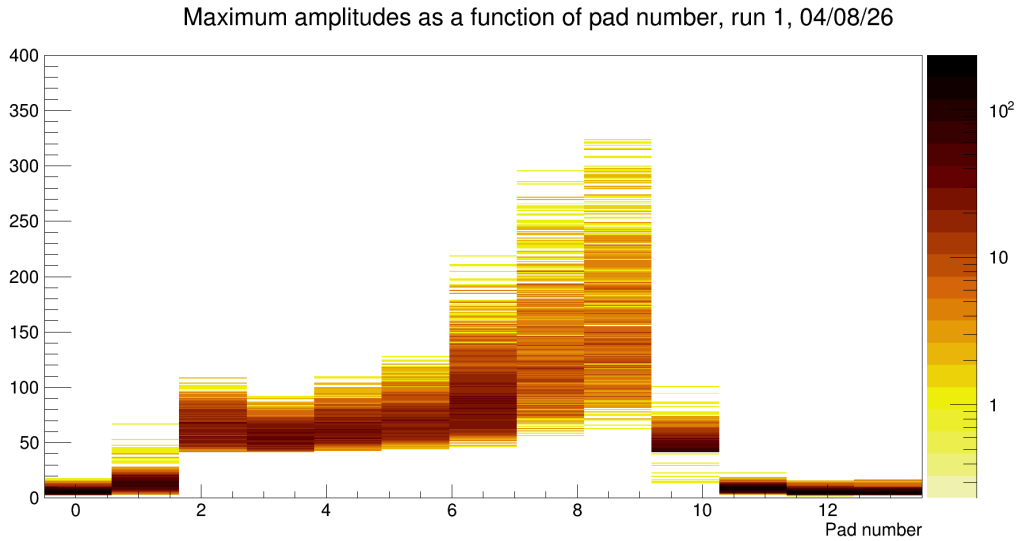


Figure 3.8: Pulse height as a function of electronic channel number for multiplicity-10 events, Run 1 (4 August 2026).

3.4 Older Ar:CO₂ data

I was also advised to examine an older dataset from 14 July 2021 in which the preamplifiers were known to be correctly connected. No comparable anomaly was observed, but the data were acquired at 500 mbar with a ⁵⁵Fe X-ray source, which produces localised interactions rather than extended alpha-particle tracks; the dataset therefore provided only qualitative

information about the pad and electronics response and was not a direct test of the track-shape anomaly [1]. I also applied the baseline-resolution comparison of Sec. 2.4 to this dataset, obtaining results similar to those obtained previously.

3.5 Hypothesis evaluation and rejection

Taken together, these observations suggested that the explanation for the observed anomaly was associated with the setup not yet having reached equilibrium in the first run, so that it progressively disappeared in the following runs. Furthermore, since the anomaly happened only at pad 2, I supposed that the source of the problem could have been near that region. In fact, pad 1 could also have been affected, but it may not have been visible because the amplitude in pad 2 was higher. Given all of this, I formulated and tested several additional hypotheses.

3.5.1 Gas inlet

A larger recorded pulse amplitude can result from a larger charge arriving at the corresponding pad. This motivated the hypothesis that a transient non-uniformity in the gas density near the inlet could have increased the ionisation, thus charge, produced in the region corresponding to the early pads. I rejected this hypothesis because the chamber was filled statically, pressure was allowed to equilibrate before acquisition, and the gas inlet (leftmost stainless steel line connected to the upper plate of the chamber in Fig. 1.2) was sufficiently separated from the relevant active region to make a measurable local effect unlikely.

3.5.2 Collimator deterioration

After seeing double pulses at the start of a couple of events, I wondered whether the collimator could be contributing to the effect, even if most anomalous events did not have it. Specifically, the unexpectedly large population of high-angle tracks appeared inconsistent with the restricted angular acceptance expected from a collimator. This led me to the hypothesis that the collimator could have deteriorated over time.

To test this, I drew normalised² histograms for the multiplicity of all events for both the new dataset and data acquired five years earlier. Results are illustrated in Fig. 3.9. The different pressures shift the absolute multiplicity distributions, so this comparison is only qualitative. Nevertheless, the shapes of the histograms indicated that the newer dataset contained a substantially larger fraction of low-multiplicity events than the older one. This was consistent with a larger fraction of high-angle alpha tracks, hinting at an issue with the collimator.

After discussing this with the group, it was confirmed that, rather than having deteriorated, no collimator was installed for the newer measurements, whereas the older dataset had been acquired with a collimator.

²Since the numbers of recorded events differed, normalising them helped visualise the differences at the same scale.

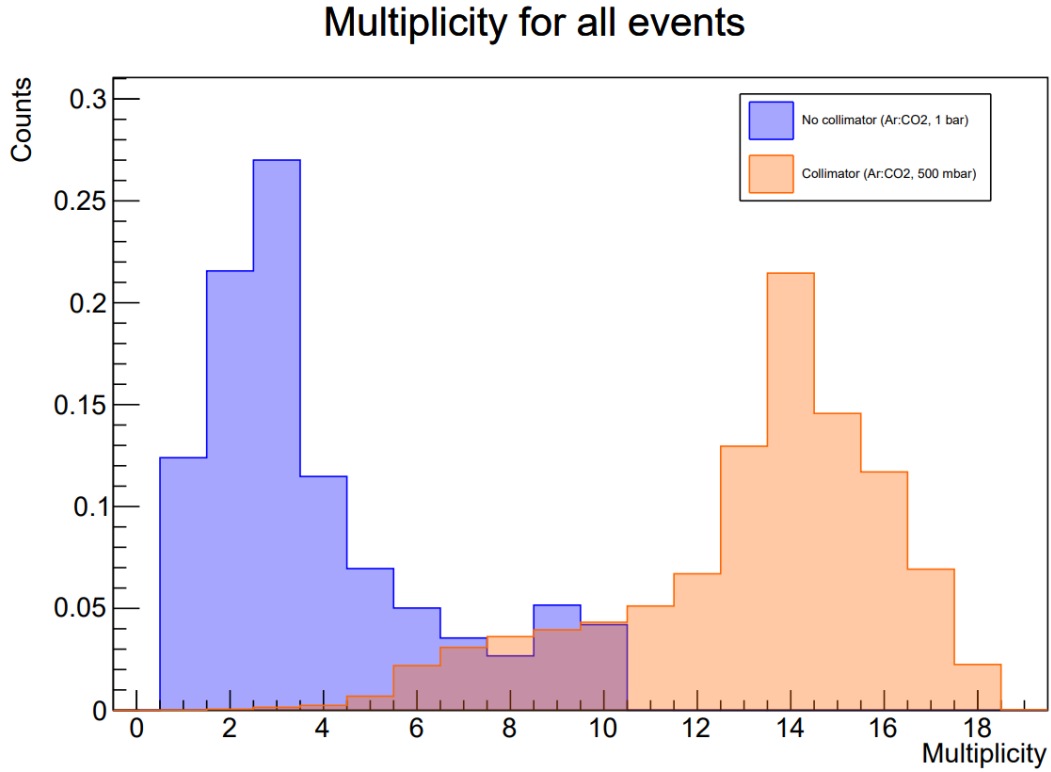


Figure 3.9: Normalised multiplicity histograms including all events from a newer and older run.

Although this only explained pulse pile-up at the start of some events and not the original anomaly I was studying, it is important for understanding the interpretation developed in the following section.

3.5.3 μ -RWELL charge-up

After discussing my findings and hypotheses with the group, a promising explanation arose. When a bias voltage is set to the detector and the chamber, the dielectric polyimide polarises under the applied electric field, producing a time-dependent modification of the local field distribution. This causes a space-charge polarisation inside the material, an effect known as polarisation charge-up [11, 12]. Moreover, electrons originating from ionisations start drifting towards the μ -RWELL, entering the holes and undergoing avalanche multiplication. Part of the avalanche charge can be deposited on the exposed polyimide surfaces. Because of the dielectric nature of the material, this deposited charge cannot flow away immediately and starts accumulating. This is known as radiation charge-up. Both phenomena can increase or decrease the local field and therefore modify the effective gain [11, 12]. According to previous studies with our setup, charge-up effects were expected to produce a net increase in the detected amplitudes [1]. In previous test-chamber measurements, the time required to reach a stable gain with an alpha source was of the order of minutes in both Ar- and He-based mixtures, substantially shorter than for low-rate ^{55}Fe irradiation because of the much larger primary-ionisation rate [1].

The charge-up rate depends on how frequently avalanche charge is deposited in the μ -RWELL region corresponding to a given pad [11, 12, 1]. Since the earlier pads were

closer to the source, the region of the μ -RWELL below them was being exposed to many low-multiplicity events, in addition to the high-multiplicity ones that the later pads were also receiving. This could then explain why the earlier regions charged more rapidly than the later ones, producing a transient non-uniformity in the measured amplitudes, which decreased as the remaining detector regions approached equilibrium.

To test this, I looked at how amplitudes evolve over time at any given pad. After investigating and trying different methods to quantify this, I divided all the events from a run into smaller blocks and plotted the mean amplitude recorded in each one of them, the block number serving as a proxy for time here, under the assumption that the event rate remained approximately constant during the run. A graph worked better in this case because the statistics within individual blocks were insufficient for stable histogram-based estimates. Fig. 3.10 shows the result of that for pad 2, where it can be seen how the mean amplitude increases over time, a behaviour consistent with charge-up.

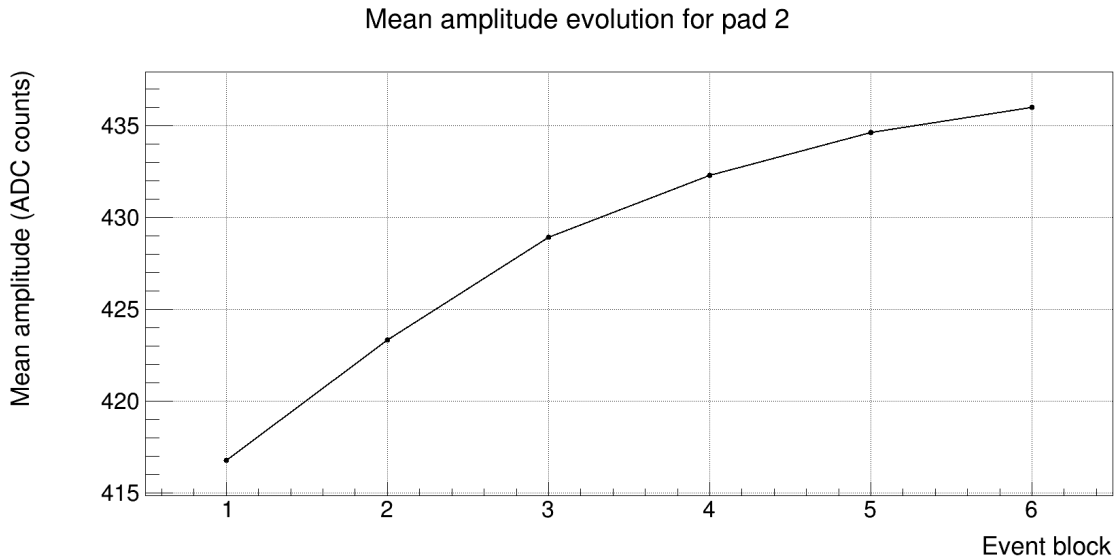


Figure 3.10: Mean pulse amplitude recorded in each block of 27000 events as a function of event block for Run 1 (8 July 2026), using only multiplicity-9 events.

I then compared that rate with the other pads and plotted them all at once, as seen in Fig. 3.11. This result shows that later pads exhibit smaller positive slopes than the earlier ones, indicating slower charge-up, as expected. I had, however, to look for a more direct and reliable way to quantify their different charging rates in order to test the hypothesis quantitatively.

Taking the difference between the last and first mean amplitude was not a reliable way to do it, nevertheless. I therefore fitted the graphs from Fig. 3.11 to a straight line (see red lines) and calculated the slopes. Since the magnitude of the amplitudes is different for each pad, I also divided that slope by the mean amplitude of all events to get the relative rate of increase in amplitude over time.

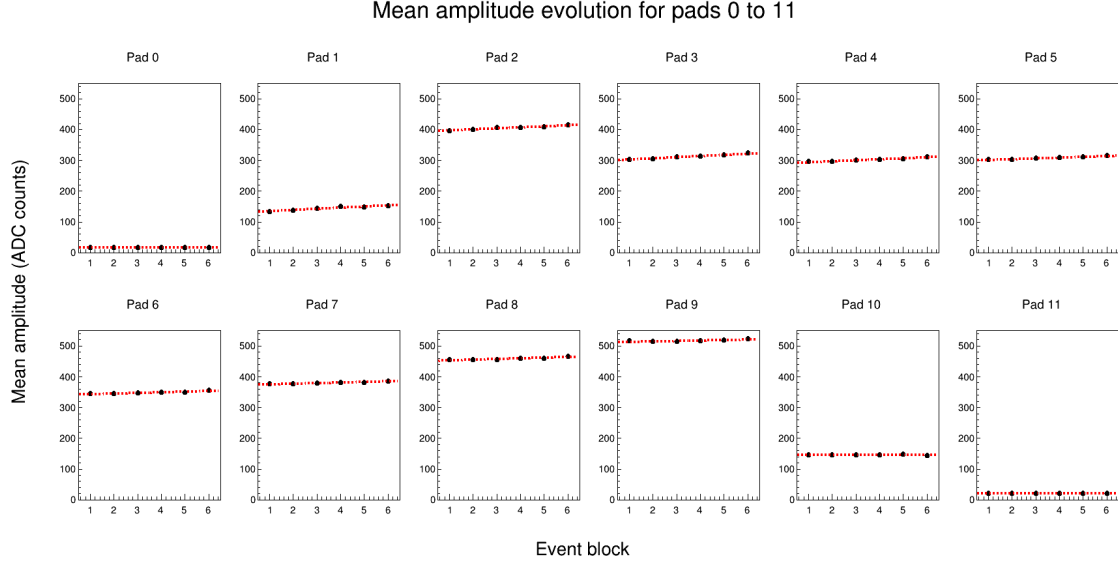


Figure 3.11: Mean maximum amplitude recorded in each block of 27000 events as a function of event block for Run 1 (8 July 2026), using only multiplicity-10 events.

The results consistently showed a trend towards charging rate decreasing as pad number increases. Pad 2, however, always showed slower charging rate than pad 3, even when varying analysis variables such as the block size or block number. I then included statistical uncertainties to quantify the precision of the estimated trends. Uncertainties for each block j in the graphs of the mean amplitudes, $\sigma_{\bar{A},j}$, were calculated as the standard error of the mean amplitude:

$$\sigma_{A,j} = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (\bar{A}_j - A_i)^2} \quad (3.1)$$

$$\sigma_{\bar{A},j} = \frac{\sigma_{A,j}}{\sqrt{N}} \quad (3.2)$$

Here, N is the number of used events of the block³ and the quantity $\sigma_{A,j}$ is the standard deviation of the amplitudes in that block. The resulting relative charging rate estimates are shown in Fig. 3.12, and present a similar trend to that described without uncertainties.

A likely explanation for this appeared to be that pad 2 was already approaching its charging-up plateau. Fig. 3.10 visually shows how the slope of its amplitude evolution decreases over time. In fact, Fig. 3.11 shows how mean amplitudes in pad 2 were already higher than in pad 3 for the very first block, indicating that it was already substantially charged when the run started. This is consistent with pad 2 recording higher mean pulse amplitudes than its neighbours, as initially seen in Fig. 3.2.

³Because I am applying a multiplicity cut here, not all events from each block are actually being used, but are instead a pool to select from.

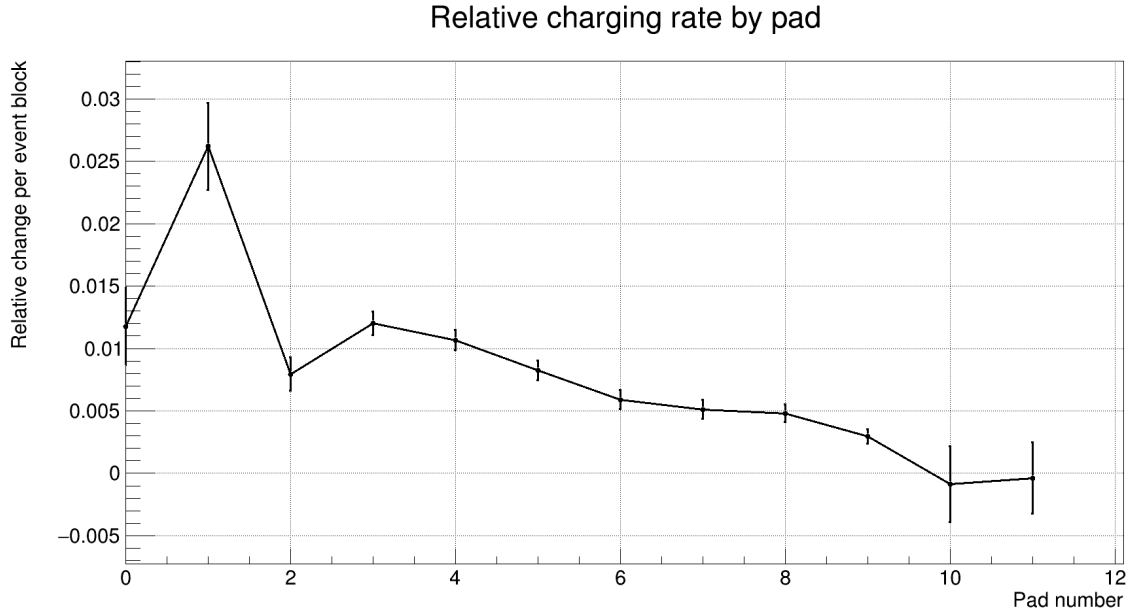


Figure 3.12: Approximate relative charging rate as a function of electronic channel number for Run 1 (8 July 2026), using only multiplicity-10 events.

Following up with the interpretation of Fig. 3.12, pad 0 did not always record a pulse because of the position of the source. Pad 1 did, while pad 2 corresponded to the end of many low-multiplicity tracks, where the alpha stopping power is larger near the Bragg region. Lastly, even the longest alpha tracks did not reach pads 10 and 11. These endpoint conditions of the earliest and latest pads can reduce the number of accepted pulse events and/or increase the spread of the recorded amplitudes, producing larger uncertainties. The remaining pads show a more uniform trend.

Finally, I plotted the same graph for the next runs to check that the detector was closer to equilibrium by then. Indeed, Fig. 3.13 supports that statement, indicating a relative charge-up effect over all pads substantially lower than for the first run.

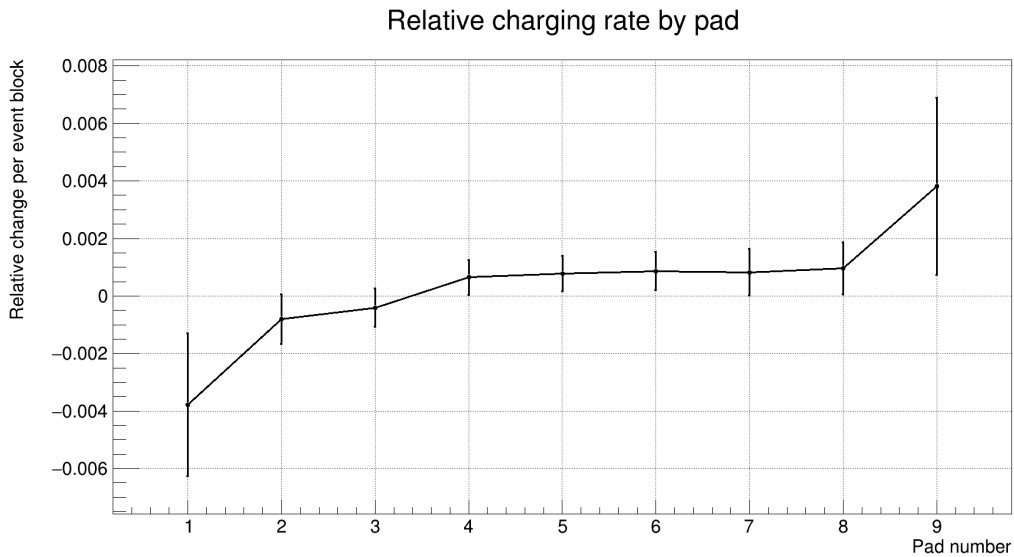


Figure 3.13: Approximate relative charging rate as a function of electronic channel number for Run 2 (8 July 2026), using only multiplicity-10 events.

Going back to Fig. 3.3, one channel can still record higher amplitudes than the next even after the detector approaches equilibrium. The main explanation considered here is the different local irradiation rate: the region closer to the source receives more low-multiplicity events and can therefore reach a different equilibrium charging state than later regions. Residual pad-to-pad differences may also receive contributions from μ -RWELL structural non-uniformities and preamplifier-channel response variations [1]. This highlights the importance of collimation in controlling the irradiation profile.

Chapter 4

Conclusions and future work

4.1 Conclusions

This report investigated several characteristics of the recorded detector waveforms. On the one hand, increasing the drift field inside the chamber was observed to increase the detected amplitudes by a small amount, consistent with improved primary-charge collection through reduced recombination and/or changes in electron collection efficiency.

On the other hand, I have roughly estimated how different calculations of the baseline of a pulse amplitude can affect results. Specifically, I estimated a 3.43% difference in the approximate relative amplitude resolution between using only the first sample of the event as the baseline and taking the average of the first 90 recordings. This change was small relative to the other limitations of the analysis.

Furthermore, I have developed a new system to identify and visualise pulse pile-up, as well as candidate background interactions.

A central focus of the project has been an anomaly found at the start of the Bragg curve for the alpha particles in this setup. After studying its behaviour and assessing different hypotheses, the observations were found to be consistent with a μ -RWELL charge-up effect. The absence of a collimator produced a non-uniform irradiation profile that supported the observed pad-dependent charging behaviour.

A key conclusion of this work is the importance of verifying the operating conditions of the detector before data acquisition. This reduces the risk of systematic errors and avoids unnecessary analysis effort.

Another important takeaway is that preprocessing and restructuring the raw data substantially reduced the computational cost of the subsequent analysis. In that context, I have also developed analysis code to do just that, which may be reused in future analyses by the group.

4.2 Improvement suggestions

Future measurements of this type should employ a collimated source. In this case, the uncollimated configuration was required for separate track-reconstruction studies; future amplitude studies should use a configuration matched to their specific objectives.

Once that is implemented, a more dedicated charge-up study should be performed with

this setup. This would help determine how long it takes for the entire detector to reach equilibrium before data acquisition.

The data taken with swapped preamplifiers could also be examined more in depth, analysing them from a charge-up perspective. The goal of this would be to estimate how much this effect influenced those runs, allowing us to subsequently quantify the difference in gain between the two preamplifiers.

A validated channel-by-channel pulser calibration should be implemented and applied before quantitative comparisons between channels.

Event-selection criteria could also be improved. Tracks extending only slightly into a tenth pad are nevertheless classified as multiplicity-10 events and may have slightly different characteristics. An alternative is to operate at lower gas pressure or use mixtures in which the particle range is longer, although this would also change the energy deposited per pad, electron transport and detector-gain requirements [2].

Moreover, the preliminary resolution comparison presented in section 2.4 could also be carried out more rigorously if necessary, for example, by using higher-statistics datasets, refining the event-selection criteria and accounting for correlations between variables.

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